Commissioning-Calibrated GP-HOCBF Safety Filtering for Ultra-Supercritical Boiler-Turbine Control under Model Mismatch

Abstract

Ultra-supercritical power-unit coordinated control must enforce pressure, separator-enthalpy, and power limits even when changing coal quality, heat absorption, and process dynamics invalidate the nominal model. Under this mismatch, a command accepted by the coordinated controller can still drive constrained variables toward their limits. This paper proposes RoCBF-SF, a commissioning-calibrated Gaussian-process high-order control barrier function (GP-HOCBF) safety filter placed between an existing controller and the actuators. Three GP models correct drift residuals, posterior uncertainty is propagated through the HOCBF chain, and a one-step quadratic program projects each proposed command. In a seven-state, actuator-augmented benchmark with $\Delta g = 0$, HOCBF without GP correction records 65,134 violating samples among 150,000 samples from six mismatch scenarios. The predeclared tune/test rule selects $\epsilon_{\kappa} = 0$; the resulting GP-mean filter and an NMPC reference each record 0/175,000 observed violations in the primary sweep, and the fixed filter setting records 0/50,000 violations and 0/50,000 QP rejections on held-out seeds. By contrast, the implemented full-margin endpoint rejects 149,800/150,000 mismatch-scenario QPs under the tested actuator limits. Plant evidence from a 660 MW ultra-supercritical unit provides a separate operating-context cohort, a native-5 s matched response pair, and 4320 confirmed high-load controller-export records. In the matched pair, pressure-error standard deviation decreases from 0.375 to 0.156 MPa. The controller exports cover 480.7–629.7 MW: two windows retain full-QP operation, one records 43 guarded reduced-QP recoveries, and the maximum total task time is 28.665 ms against a 1000 ms deadline. These results support a deployable safety-filter workflow while distinguishing finite-sample calibration, observational plant response, controller execution, and the conditions required for a formal certificate.

Keywords: Ultra-supercritical boiler-turbine control; model mismatch; safety filter; high-order control barrier function; Gaussian process residual learning; robustness-margin calibration

1 Introduction

1.1 Industrial Measurement and Control Context

During load following and deep peak shaving, ultra-supercritical power units operate close to equipment protection limits. Their boiler-turbine coordinated control systems must keep pressure, separator enthalpy, and power within bounds while coordinating fuel, feedwater, and turbine-valve actuation. For this plant class, constraint handling is a coordinated control problem, not only a nominal tracking problem.

The controller design model is imperfect in service. Coal quality varies across shipments, heat absorption changes as heat-transfer surfaces foul, valve and fuel-delivery characteristics drift, and unmodeled coupling becomes more visible during load following. These effects create persistent mismatch between the controller model and the true process response. A controller that appears feasible under the nominal model, whether PID, LQR, MPC, or a learned policy, can still drive constrained variables toward a pressure, enthalpy, or power violation.

Recent applied-control studies similarly report disturbance response and computational update burden alongside tracking accuracy when plant response departs from the nominal design[1,2]. We adopt the same discipline for boiler-turbine model mismatch by treating safety-margin selection as a commissioning task for a downstream safety-filter overlay.

1.2 Safety-Filter View

We study a downstream safety filter, not a replacement for the plant controller. The filter receives current plant measurements and the upstream controller's proposed command, checks that command against process safety constraints, and returns the nearest admissible command. This placement suits industrial control practice because it separates tracking from protection: the existing controller handles load following and economic operation, while the safety layer enforces the final operating constraints.

We implement the projection as a low-latency quadratic program (QP) whose rows are derived from control barrier functions (CBFs)[3]. Relative degree must be defined with respect to the controller command rather than an unmeasured actuator state. After the fuel, feedwater, and turbine-valve dynamics are included, each pressure, separator-enthalpy, and power limit used here has command-level relative degree two. High-order CBFs (HOCBFs)[4] are therefore needed to preserve the constraint dynamics. A first-order CBF baseline would incorrectly assume that the commanded input appears in the first barrier derivative; that structural error is distinct from the plant-model mismatch studied below.

1.3 Technical Gap: Uncertainty Propagation in High-Relative-Degree Constraints

Gaussian processes (GPs) provide a natural way to learn model residuals from commissioning or operation data while quantifying posterior uncertainty[5,6]. Existing GP-CBF methods usually apply the uncertainty bound to first-order barrier constraints[7–9], while recent GP-HOCBF work extends learning-based safety to high-order constraints through chance-constraint or convex reformulations[10,11]. These studies establish useful theory, but they do not by themselves resolve the commissioning problem faced by actuator-limited boiler-turbine loops.

For boiler-turbine control, the difficulty is structural. The selected pressure, separator-enthalpy, and power limits are command-level relative-degree-two constraints, and residual uncertainty propagates through the HOCBF recursion rather than appearing only at the final constraint row. Applying the full implemented margin can over-tighten the online QP, reduce actuator authority, or trigger fallback behavior. Applying no margin can also be unsafe when the perturbation is state-dependent and amplifies along the trajectory. The unresolved question is therefore not only how to construct a GP-HOCBF margin, but how much of that margin should be applied in real time for the residual structure observed during commissioning.

RoCBF-SF addresses this question by combining GP residual mean correction, compositional uncertainty propagation through the HOCBF ψ-chain, QP projection, and a robustness factor calibrated from commissioning residuals. The scientific contribution is the safety-filter overlay and its calibration procedure under measured model mismatch, rather than a new upstream controller.

1.4 Contributions

We introduce RoCBF-SF, a commissioning-calibrated GP-HOCBF safety filter for ultra-supercritical boiler-turbine control. The paper makes four contributions to industrial measurement, constrained control, and real-time deployment:

1. **Downstream safety-filter architecture for boiler-turbine process control.** RoCBF-SF preserves the upstream controller while enforcing command-level relative-degree-two pressure, separator-enthalpy, and power constraints obtained from an actuator-augmented process model.
2. **Compositional uncertainty propagation through the HOCBF chain.** The method propagates GP posterior residual uncertainty through each HOCBF ψ-level and embeds the resulting state-dependent margin in a real-time QP projection.
3. **Commissioning-calibrated robustness factor.** The scalar factor ϵ_{κ} interpolates between mean-corrected operation and the full implemented uncertainty margin, with diagnostics for selecting this factor under additive and state-dependent mismatch.
4. **Simulation validation and measured-operation evidence.** The approach is evaluated on an actuator-augmented boiler-turbine benchmark, while bounded production historian records and controller-log exports from a 660 MW ultra-supercritical unit provide response, execution, feasibility, and latency evidence under measured operation.

The validation is organized around this applied process-control logic. Controlled simulations first isolate the model-mismatch mechanism and show when GP mean correction and ϵ_{κ} calibration are needed. Plant historian records then compare pre- and post-retrofit response under matched load movement, and controller-log exports verify the online execution path across repeated post-retrofit windows. Together, these evidence sources connect the formal safety-filter mechanism to plant measurements while keeping the upstream controller outside the proposed method.

2 Related Work

2.1 Boiler–Turbine Coordinated Control Under Model Mismatch

Boiler–turbine units are multivariable, nonlinear, delayed, and strongly coupled process-control systems. The Astrom–Bell benchmark made this class of dynamics a standard control problem[12]. Later coordinated control studies addressed boiler–turbine load following, pressure regulation, and actuator coordination through fuzzy reasoning, internal-model or linear-control structures, supercritical-unit coordinated control, and constrained predictive control[13–15]. These studies establish the boiler–turbine loop as a constrained coordinated control problem rather than a single-input tracking problem.

More recent boiler–turbine work has pushed this line toward constrained predictive control, hierarchical optimization, and operational flexibility. Constrained predictive schemes address thermal limits during power-plant coordinate control[15], and recent boiler–turbine MPC formulations emphasize feasibility, input limits, and load-response capability[16]. Coal-fired flexibility studies further show why load-following operation is exposed to coal-quality variation, heat-absorption drift, and actuator coordination limits[17]. The remaining gap for the present paper is architectural: most prior boiler–turbine controllers replace or redesign the coordinated controller itself, whereas an industrial retrofit often needs a downstream safety layer that can be inserted after the existing controller and verified through plant measurements.

2.2 Constrained Process Control: MPC, Robust MPC, and Safety Filters

MPC remains the dominant constrained-control methodology in process industries[18,19], and offset-free MPC addresses steady-state model mismatch[20]. Robust and stochastic MPC extend this framework by tightening constraints against bounded or probabilistic uncertainty[21,22], and cautious MPC incorporates GP uncertainty into the prediction horizon[23]. These approaches provide a natural benchmark for boiler–turbine constraint handling, but their online optimization, horizon modelling, and controller-replacement assumptions differ from a low-latency retrofit filter.

A safety filter takes a different architectural position. Rather than re-solving the full tracking problem, it modifies a proposed command only when that command threatens the constraint set. This role also matches applied control reporting, where pressure regulation is judged through delay and disturbance response[1] and update burden is reported with tracking performance[2]. RoCBF-SF uses this one-step filtering role to keep the upstream boiler–turbine controller outside the proposed method.

2.3 CBF/HOCBF Under Uncertainty and Feasibility Limits

Control barrier functions enforce forward invariance through QP-based control projection[3,24]. HOCBFs extend this idea to high-relative-degree constraints[4]. They are necessary here because actuator dynamics delay the effect of each controller command, making the selected pressure, enthalpy, and power barriers relative degree two at the command interface. Robust and adaptive CBF formulations add disturbance or parameter margins[25–27], but actuator bounds and multiple barrier rows can still make the online CBF-QP infeasible. Feasibility-aware CBF work therefore optimizes barrier decay rates or modifies the safety program to maintain pointwise feasibility under input constraints[28].

The sampled-data implementation adds another boundary. Discrete CBFs translate barrier conditions to discrete-time systems[29], sampled-data CBFs address the gap between continuous-time design and discrete-time implementation[30], and recent stochastic discrete-time CBF work studies robustness under uncertainty[31]. These results motivate explicit reporting of sample time, zero-order hold, feasibility relaxation, fallback logic, and inter-sample checks. The present work gives a conditional continuous-time GP-HOCBF result and treats the sampled implementation as an engineering boundary that must be evaluated by offline benchmark tests and controller logs exported from the unit.

2.4 GP Residual Learning and Commissioning-Calibrated Margins

GPs provide nonparametric residual learning with uncertainty quantification[5,6]. Uniform GP error bounds support safe-control certificates under data-limited residual models[32], and Bayesian measurement studies similarly use posterior uncertainty to quantify calibration limits from limited data[33]. GP-CBF methods combine this residual-learning logic with safety filters. Cheng et al.[7] use GPs to support safe reinforcement learning, Fisac et al.[8] develop a learning-based safety framework for uncertain systems, and Castañeda et al.[9] analyze pointwise feasibility for GP-based CBF-CLF optimization under model uncertainty.

For high-order constraints, the comparison is narrower. Aali and Liu[10] integrate GPs with HOCBFs by converting chance constraints into a second-order cone safety filter and analyzing feasibility. Zhang et al.[11] develop a real-time sparse-GP HOCBF filter for high-order robotic systems. These methods show that GP uncertainty can support HOCBF safety filters, but they do not target command-level relative-degree-two boiler–turbine constraints, DCS deployment, or commissioning diagnostics based on plant historian and controller-log records. Existing robust CBF and GP-CBF formulations also tend to interpret the uncertainty margin as a certificate-driven tightening term. In actuator-limited process-control loops, however, feasibility, fallback activation, and retained control authority are themselves safety-relevant. This leaves a commissioning question: how should residual-data coverage and replay diagnostics determine the usable fraction of the GP-HOCBF margin?

3 Commissioning-Calibrated GP-HOCBF Safety-Filter Architecture

3.1 Safety-Filter Workflow and Implementation Settings

Online operation has three steps: evaluate the GP residual model, build the command-level relative-degree-two HOCBF rows, and solve one QP to project the proposed deviation command. Offline commissioning supplies the residual data, GP coverage checks, and scalar margin factor ϵ_κ. The theorem states the conditions under which the full implemented-margin endpoint yields continuous-time forward invariance. Section 4 then evaluates how much of that margin should be applied under each mismatch regime.

Figure 1 summarizes the overlay architecture. The upstream controller produces a deviation command v_{prop} around the LQR-stabilized equilibrium controller, and the safety filter returns the nearest admissible command v^*. The normalized deviation command satisfies $v \in \mathcal{V} = [-1,1]^3$. With the physical per-cycle command scales $S_u = \text{diag}(10,40,1)$, the total actuator command is

$$u_k = u_0 + K(x_0 - x_k) + S_u v_k^*, \tag{1}$$

where $u = [u_B, D_{fw}, u_t]^\top$ denotes coal-feed, feedwater-flow, and turbine-valve commands. The HOCBF and QP are formulated in normalized deviation coordinates. Let $A_d = \exp[(A - BK)T_s]$, $A_s = (A_d - I)/T_s$, and $B_s = BS_u$. The sample-matched affine surrogate used to build the benchmark HOCBF rows is

$$\dot{\xi} = A_s\xi + B_s v + \Delta f_\xi(\xi). \tag{2}$$

Its forward-Euler update is $\xi_{k+1} = A_d\xi_k + T_s B_s v_k + T_s \Delta f_\xi(\xi_k)$, which is the controller-instant transition used in the numerical rollouts. This construction preserves the continuous command-level relative degree induced by the actuator states. It is a benchmark surrogate rather than a sampled-data safety certificate. Table 1 lists the benchmark constraint definitions alongside the principal GP, QP, fallback, and runtime settings. Field-specific direct bounds are reported with the plant-controller evidence in Section 4.5.

Table 1. Benchmark constraint definitions and principal implementation settings used by the RoCBF-SF safety filter. Field-specific direct bounds are reported separately with the plant-controller evidence.

Item	Setting	Role in the filter
Benchmark state	$x = [r_B, p_m, h_m, N_e, \tau_f, D_{fw}$	Five process states from the reference model plus measured feedwater and valve actuator states.
GP input/output	$x_k^{\text{GP}} = [p_{m,k}, h_{m,k}, N_{e,k}]^\top$ and three residual rates	Constrained outputs used by the residual GPs; all seven states and three proposed commands remain inputs to the nominal one-step predictor.

Item	Setting	Role in the filter
Manipulated inputs	$u=[u_B, D_{fw}, u_t]^{\mathrm{T}}$; $S_u=\mathrm{diag}(10,40,1)$	Physical commands are reconstructed from normalized $v\in[-1,1]^3$.
Benchmark pressure constraint	$p_{\mathrm{st}}(x)\in[13,24]$ MPa; relative degree $m=2$	High-order pressure CBF with two ψ-levels.
Benchmark enthalpy constraint	Range 2670–2830 kJ/kg; relative degree $m=2$	Actuator-augmented heat-absorption safety constraint.
Benchmark power constraint	Band $N_{\mathrm{ref}}\pm 50$ MW; relative degree $m=2$	Actuator-augmented generator-output constraint around the dispatch target.
Sampling and hold	Controller period $T_s=1$ s; zero-order hold	Online QP solved once per controller cycle; plant-validation exports use 5 s record spacing.
GP calibration	Three 500-point Matérn-5/2 residual GPs with frozen input/output standardization	Learns pressure, enthalpy, and power residual-rate means and posterior uncertainties; 200 time-isolated points are retained for validation.
QP implementation	qpax, row scaling, box/rate constraints, 10^{-6} feasibility tolerance	Projects v_{prop} to the nearest admissible v^* and rejects non-converged or infeasible candidates.
Recovery and bypass	One guarded pressure_low reduced-QP retry; otherwise live DCS command	The guarded retry is outside the full-row certificate; actuator rows are never removed.
Runtime deadline	Controller period and deployed task deadline both 1000 ms	Field exports report QP and total-task timing against the deployed deadline.
Plant controller platform	AMD EPYC Embedded 4565P; 64 GB (2×32 GB) DDR5-5600 ECC; openEuler 24.03 LTS	Deployed controller compute platform; data exchange with the DCS uses Modbus TCP.
Plant software stack	Python 3.11; qpax 0.1.3; SciPy 1.17.1; Kubernetes	qpax provides the online QP solve; Kubernetes manages the deployed workload.

Online filtering algorithm. At each sampling instant, RoCBF-SF performs the following operations:

1. Read x_k and the proposed deviation action $v_{\mathrm{prop},k}$.
2. Evaluate $\mu_{\mathrm{GP}}(x_k)$ and $\sigma_{\mathrm{GP}}(x_k)$.
3. Build the mean-corrected command-level relative-degree-two HOCBF constraints.
4. Propagate GP uncertainty through the ψ-chain to compute $\sigma_{\mathrm{total}}(x_k)$.
5. Solve the real-time QP
$$v_k^*=\arg\min_v \| v-v_{\mathrm{prop},k}\|_2^2 \quad \mathrm{s.t.} \quad A(x_k)v\le b(x_k)-\epsilon_\kappa\sigma_{\mathrm{total}}(x_k), \quad v\in\mathcal{V}, \tag{3}$$
where $\mathcal{V}$ is the deviation-input box induced by actuator limits. Row scaling is applied before the numerical solve and does not change the feasible set.
6. Apply $u_k=u_0+K(x_0-x_k)+S_u v_k^*$ and log intervention, feasibility, and margin

diagnostics.

Notation summary. The benchmark state is $x = [r_B, p_m, h_m, N_e, \tau_f, D_{fw}, u_t]^\top$, and $\xi = x - x_0$ is the deviation state. The physical command is u, while v is the normalized command deviation optimized by the safety filter. The upstream controller proposes $v_{\mathrm{prop},k}$, the QP returns v_k^*, and Eq. (1) maps v_k^* back to the physical command. The GP-mean-corrected HOCBF rows are written as $A(x)v \leq b(x)$; $A_0(x)v \leq b_0(x)$ is reserved for the no-GP nominal-model ablation. The term $\sigma_{\mathrm{total}}(x)$ is the compositional GP-HOCBF margin, and ϵ_κ is the commissioning factor multiplying that margin.

Both the benchmark and plant implementations use the measured-output vector

$$x_k^{\mathrm{GP}} = [p_{m,k}, h_{m,k}, N_{e,k}]^\top. \tag{4}$$

Three independent GPs model the corresponding one-step nominal-model residual rates. The benchmark target is $(x_{k+1,J_{\mathrm{GP}}} - \hat{x}^{\mathrm{nom}}_{k+1|k,J_{\mathrm{GP}}})/T_s$, where $J_{\mathrm{GP}} = \{p_m, h_m, N_e\}$. The full nominal predictor uses all seven states and all three command deviations; commands are therefore represented in the nominal prediction but are not direct kernel inputs.

Operating-point interpretation. RoCBF-SF denotes the complete safety-filter design in Fig. 1: correct-relative-degree HOCBF rows, GP residual mean correction, compositional uncertainty propagation, QP projection, and tunable ϵ_κ. The upstream controller may be classical or learned; the experiments evaluate the downstream filter contribution rather than the upstream controller. Experimental rows labelled GP-HOCBF and RoCBF-SF are fixed operating points or ablations. Thus, $\epsilon_\kappa = 0$ tests mean correction alone, $\epsilon_\kappa = 1$ tests the full implemented margin, and Section 4.3 selects and evaluates the calibrated benchmark operating point.

3.2 Mismatch Model and HOCBF Constraints

Consider a control-affine system with model mismatch:

$$\dot{x} = \underbrace{f_0(x) + g_0(x)u}_{\text{nominal}} + \underbrace{\Delta f(x)}_{\text{residual}}, \quad x \in \mathbb{R}^n,\ u \in \mathcal{U} \subset \mathbb{R}^m \tag{5}$$

where Δf represents unmodeled dynamics, parameter drift, and external disturbances. The formal certificate assumes a known input matrix ($\Delta g = 0$), so the GP learns drift residuals rather than uniform actuator-gain uncertainty.

The benchmark pressure output is $p_{\mathrm{st}}(x) = p_m - 0.13 p_m^{0.882}$. The six scalar safety barriers are

$$\begin{aligned} h_{p,L}(x) &= p_{\mathrm{st}}(x) - 13, \quad & h_{p,U}(x) &= 24 - p_{\mathrm{st}}(x), \\ h_{h,L}(x) &= h_m - 2670, \quad & h_{h,U}(x) &= 2830 - h_m, \\ h_{N,L}(x) &= N_e - (N_{\mathrm{ref}} - 50), \quad & h_{N,U}(x) &= N_{\mathrm{ref}} + 50 - N_e. \end{aligned} \tag{6}$$

The five process states of the reference model are augmented with first-order feedwater and turbine-valve states, identified as 15 and 6 s from two unsaturated controller-export windows; the existing 30 s fuel-transport state is retained. With controller commands as inputs, all six barriers have relative degree two. This removes the direct-command artifact that would otherwise make separator enthalpy and power relative degree one.

For a constraint $h(x) \geq 0$ with relative degree m, the HOCBF ψ-chain is:

$$\psi_0 = h(x), \quad \psi_i = \dot{\psi}_{i-1} + \alpha_i(\psi_{i-1}), \quad i = 1, \ldots, m \tag{7}$$

For the six barriers indexed by c, with relative degree m_c, define the recursive HOCBF admissible set under the true dynamics as

$$\mathcal{C}_{c,i} = \{x \in \mathcal{X}_{\text{op}} : \psi_{c,i}^{\text{true}}(x) \geq 0\}, \quad i = 0, \ldots, m_c - 1, \qquad \mathcal{C} = \bigcap_{c=1}^{6} \bigcap_{i=0}^{m_c-1} \mathcal{C}_{c,i}. \tag{8}$$

With linear $\alpha_i(r) = k_i r$, substituting Eq. (1) into the stabilized model gives a top-level affine row. For the GP-mean-corrected drift $\hat{f} = f_0 + \mu_{\text{GP}}$, this row is defined by $\hat{\psi}_{c,m_c}^{0}(x, v) = b_c(x) - A_c(x)v$ and is stacked as $A(x)v \leq b(x)$. Applying the same construction to f_0 without GP correction gives $A_0(x)v \leq b_0(x)$, which is used only for the no-GP ablation. This convention keeps all six command-level relative-degree-two rows and the normalized QP decision variable v in the same coordinates.

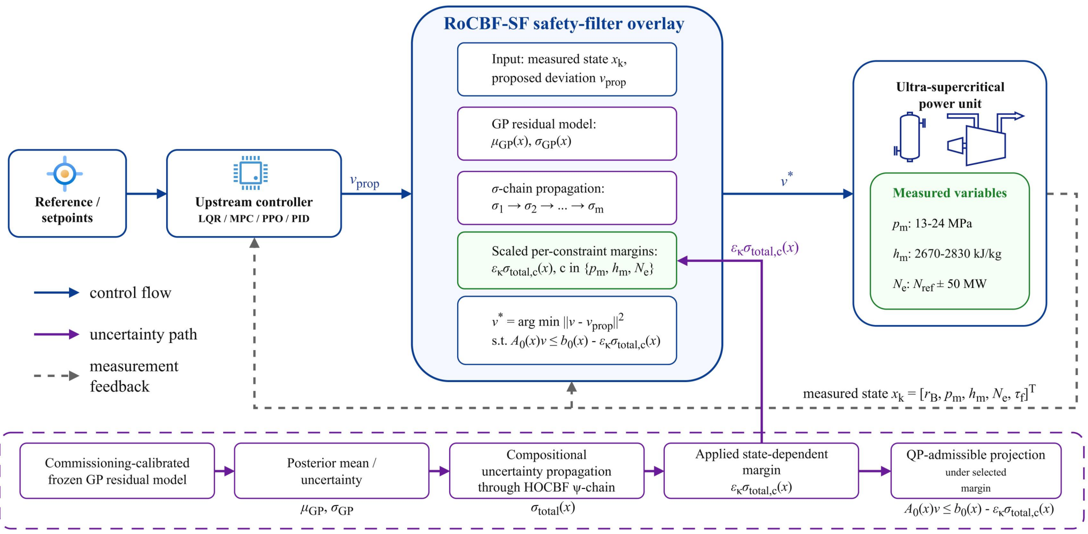

Figure 1. RoCBF-SF safety-filtering architecture. The upstream controller (LQR, MPC, PPO, or learned policy) proposes a normalized deviation command v_{prop}. The filter evaluates that command against current plant measurements and the GP residual model, then projects it to the nearest QP-admissible deviation command v^* under $A(x)v \leq b(x) - \epsilon_\kappa \sigma_{\text{total}}(x)$. The physical command is reconstructed with S_u by Eq. (1).

Figure 1 provides the system map for the rest of the paper. Figure 2 follows the signal path through plant response and QP intervention, and Fig. 3 checks the residual-model evidence for GP correction.

3.3 GP Residual Correction and Compositional Margin

Each modeled component of Δf_j, $j \in J_{\text{GP}}$, is represented by an independent GP with a Matérn-5/2 kernel. Kernel distances use the frozen training transform $\tilde{x}_i = (x_i - \mu_{x_i}^{\text{train}})/s_{x_i}^{\text{train}}$; the standard-deviation floor is 10^{-6} of the corresponding engineering range. Residual targets are standardized separately for each output. The same frozen transforms are used for training, validation, and inference. Given N training points, the posterior provides mean $\mu_{\text{GP},j}(x)$ and variance $\sigma_{\text{GP},j}^2(x)$.

The formal analysis conditions on the simultaneous residual event

$$|\Delta f_j(x) - \mu_{\mathrm{GP},j}(x)| \le \beta\sigma_{\mathrm{GP},j}(x), \quad j \in J_{\mathrm{GP}},\ x \in \mathcal{X}_{\mathrm{op}}, \tag{9}$$

with joint probability at least $1-\delta$. A standard RKHS sufficient multiplier includes both a residual RKHS-norm bound B_Δ and a sub-Gaussian noise scale R, for example $\beta_{\mathrm{RKHS}} = B_\Delta + R\sqrt{2(\gamma_N + 1 + \ln(q/\delta))}$ for $q = |J_{\mathrm{GP}}|$ modeled outputs[34]. The numerical studies and deployed configuration instead retain the fixed commissioning multiplier $\beta_{\mathrm{comm}} = \sqrt{2(\gamma_N + 1 + \ln(q/\delta))}$. This engineering scaling rule defines the reported posterior envelope but does not, by itself, establish Eq. (9); Theorem 1 applies only when that event is independently valid.

We define the mean-corrected drift $\hat{f} = f_0 + \mu_{\mathrm{GP}}$ and posterior residual $\Delta\hat{f} = \Delta f - \mu_{\mathrm{GP}}$. The induced perturbation is propagated level by level through the HOCBF chain rather than attached only to the final constraint row.

The perturbation at level i is $\delta_i = \psi_i - \hat{\psi}_i^0$, satisfying:

$$\begin{gathered} \delta_1 = L_{\Delta\hat{f}} h \\ \delta_i = L_{\Delta\hat{f}}\hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i)\delta_{i-1} + L_{\Delta\hat{f}}\delta_{i-1},\ \ i \ge 2 \end{gathered} \tag{10}$$

This two-line recursion separates the nominal HOCBF recursion from mismatch residuals.

The compositional margin is constructed as:

$$\begin{gathered} \sigma_1(x) = \beta \sum_j |\,\partial h/\partial x_j|\sigma_{\mathrm{GP},j}(x), \\ \sigma_i(x) = \beta \sum_j |\,\partial\hat{\psi}_{i-1}^0/\partial x_j|\sigma_{\mathrm{GP},j}(x) + (\,\|\,L_{\hat{f}}\|_{\mathrm{op}} + k_i)\sigma_{i-1}(x) + \sigma_{\mathrm{cross}}^{(i)}(x), \quad i \ge 2, \\ \sigma_{\mathrm{total}}(x) = \sigma_m(x) + \sum_{j=1}^{m-1} c_j\,\sigma_j(x) + \sigma_{\mathrm{ctrl}}(x). \end{gathered} \tag{11}$$

The three margin relations above define the quantity scaled by ϵ_κ in the online QP. For the formal result, $\|\,L_{\hat{f}}\|_{\mathrm{op}}$ is a uniform operating-set bound and $\sigma_{\mathrm{cross}}^{(i)}$ must bound $|L_{\Delta\hat{f}}\delta_{i-1}|$; Lemma S1 gives a sufficient derivative-bound construction. The numerical implementation uses the nominal stabilized drift in the uncertainty recursion and the commissioning proxy $\sigma_{\mathrm{cross,comm}}^{(i)}(x) = \beta_{\mathrm{comm}}\,\|\nabla\hat{\psi}_{i-1}^0(x)\|_2\,\|\,\sigma_{\mathrm{GP}}(x)\|_2$. In the linear stabilized benchmark, the nominal drift-Jacobian norm is global. For a nonlinear deployment, an equilibrium Jacobian is only a local engineering estimate. Neither the proxy cross term nor a local Jacobian estimate is automatically a certificate bound; Assumption 3 states the additional dominance condition required by Theorem 1. The control-coupling term is

$$\sigma_{\mathrm{ctrl}}(x) = \beta v_{\max} \sum_j \left\|\frac{\partial(L_{g_v} L_{\hat{f}}^{m-1} h)}{\partial x_j}\right\|_2 \sigma_{\mathrm{GP},j}(x), \tag{12}$$

where $g_v = g_0 S_u$ and $v_{\max} = \sup_{v \in \mathcal{V}} \| v \|_2 = \sqrt{3}$. The chain weights are $c_j = \prod_{\ell=j+1}^{m-1} (\| L_{\hat{f}} \|_{\mathrm{op}} + k_\ell)$, with the empty product equal to one; hence $c_1 = 1$ for each implemented $m = 2$ barrier. The resulting mean-corrected HOCBF constraint is $A(x)v \leq b(x) - \epsilon_\kappa \sigma_{\mathrm{total}}(x)$.

3.4 Commissioning Factor and Implementation Boundary

Assumption 1 (Operating set and relative degree). *The state remains in a compact operating set $\mathcal{X}_{\mathrm{op}} \subset \mathcal{X}$. The true and mean-corrected drifts, input vector fields, GP posterior functions, and barrier functions are locally Lipschitz and continuously differentiable to the order required by the HOCBF recursion. Relative degree is constant on $\mathcal{X}_{\mathrm{op}}$. The full-row QP has a unique solution whose feedback selection is locally Lipschitz on the certified set, so the closed-loop differential equation admits a unique Carathéodory solution.*

Assumption 2 (Drift-residual GP event). *The input matrix is exact ($\Delta g = 0$). For the $q = 3$ modeled residual outputs $J_{\mathrm{GP}} = \{p_m, h_m, N_e\}$, Eq. (9) holds simultaneously over $\mathcal{X}_{\mathrm{op}}$ with probability at least $1 - \delta$. This event may be established from valid RKHS and noise bounds or another justified uniform-error result; the commissioning multiplier β_{comm} alone is not such a proof. Residuals in the unmodeled r_B, τ_f, D_{fw}, and u_t rows are zero in the drift-only benchmark or are otherwise covered by Assumption 3.*

Assumption 3 (Compositional residual bound). *The derivative constants used in Lemma S1 are finite on $\mathcal{X}_{\mathrm{op}}$, the operator-norm and cross-term quantities are valid upper bounds rather than local proxies, and the margin definitions in Section 3.3, together with Eq. (12), satisfy $|\Delta_\psi(x, v)| \leq \sigma_{\mathrm{total}}(x)$ for every active HOCBF row and every admissible $v \in \mathcal{V}$.*

Assumption 4 (Robust-QP feasibility). *For every $x \in \mathcal{X}_{\mathrm{op}}$ considered by the certificate, there exists at least one $v \in \mathcal{V}$ satisfying $A(x)v \leq b(x) - \sigma_{\mathrm{total}}(x)$.*

Theorem 1 (Conditional probabilistic forward invariance). *Let $\Delta_\psi(x, v) = \psi_m^{\mathrm{true}}(x, v) - \widehat{\psi}_m^0(x, v)$ denote the induced top-level HOCBF residual after substituting the deviation-input model. If $x(0) \in \mathcal{C}$, then, under Assumptions 1–4, the QP in Eq. (3) with $\epsilon_\kappa = 1$ renders the set $\mathcal{C}$ in Eq. (8) forward invariant with probability at least $1 - \delta$.*

Proof. The complete proof is given in the Supplemental Material. It conditions on the simultaneous residual event, applies the derivative-bound construction in Lemma S1, invokes Assumption 3 to obtain $|\Delta_\psi(x, v)| \leq \sigma_{\mathrm{total}}(x)$, and then applies the HOCBF forward-invariance result to the tightened QP. The certificate is attached to $\epsilon_\kappa = 1$, independently valid residual and derivative bounds, and robust-QP feasibility.

Theorem 1 is a conditional worst-case statement. The margin itself does not imply feasibility; feasibility depends on safe-set geometry, actuator limits, and $\mathcal{U}$. The online factor ϵ_κ therefore has a dual interpretation: at $\epsilon_\kappa = 1$ it applies the complete implemented margin and recovers the formal bound only when Assumptions 2–4 are established; smaller values are commissioning settings selected from replay and operating-envelope diagnostics.

The robustness scaling factor ϵ_κ controls the safety-margin/control-authority trade-off:

$\epsilon_\kappa = 0$: The QP enforces $A(x)v \leq b(x)$, i.e., the GP-mean-corrected HOCBF constraint without an additional robustness margin. The separate $A_0(x)v \le b_0(x)$ notation denotes the no-GP ablation.

$\epsilon_\kappa = 1$: The QP enforces the full implemented compositional margin $\sigma_{\text{total}}(x)$. It recovers the worst-case certificate only when the residual event, derivative bounds, $\Delta g = 0$, and full-row QP feasibility required by Theorem 1 all hold.

$0 < \epsilon_\kappa < 1$: The QP enforces a fraction of the compositional margin, trading theoretical coverage for reduced QP conservatism.

The conditional probabilistic certificate is available only at $\epsilon_\kappa = 1$ when every assumption of Theorem 1 is established; setting $\epsilon_\kappa = 1$ alone is insufficient. Values $0 \le \epsilon_\kappa < 1$ are calibrated operating points evaluated by finite-sample violation, QP intervention, infeasibility, and deployment-envelope diagnostics. Section 4 evaluates the full implemented-margin endpoint in the actuator-limited benchmark and how the preferred value changes with mismatch structure.

The GP posterior includes a regularization floor $\sigma_{\text{floor}} = 10^{-4}$. Under constant perturbations, σ_{GP} and $\sigma_{\text{total}}(x)$ are nearly uniform, so the margin is floor-driven. Under S3 state-dependent coupling, σ_{GP} becomes heterogeneous and $\sigma_{\text{total}}(x)$ is data-driven. This distinction explains why additive perturbations can use $\epsilon_\kappa = 0$ in the tested regime: GP mean correction supplies the effective adjustment, while the floor-driven margin mainly adds QP conservatism.

Theorem 1 is continuous-time, whereas the implementation uses $T_s = 1$ s zero-order hold. One second is the deployed coordinated-control task period and allows the filter to evaluate each upstream command update; it is not intended to imply that the dominant boiler thermal dynamics have a 1 s time constant. The deployed task deadline is 1000 ms and the configured QP timeout is 200 ms; the observed controller-task and QP times are reported separately in Section 4.5. The 5 s spacing in the controller exports is a logging interval rather than the internal execution period. A failed original QP permits at most one reduced-QP attempt after removal of the whitelisted pressure_low row, and only when $\| G_i S_u \|_2 < 0.01$, the unscaled row right-hand side is below -10^{-6}, the directly measured pressure margin is at least 2.0 MPa, measurement quality is good, and no protection or interlock is active. Actuator box/rate rows are never removed. A failed retry, hard fault, or timeout returns the live coordinated-control command and latches bypass according to the deployment state machine. The reduced-QP recovery does not satisfy Assumption 4 for the original six-row problem, so Theorem 1 does not apply during those recovered cycles. Inter-sample forward invariance is not established by the reported controller-instant tests and remains outside the formal certificate.

For field commissioning, approximately 14 days of 1 Hz data are divided into days 1–10 for the training candidate pool, day 11 as a 24 h isolation gap, and days 12–14 for validation. Five fixed load strata cover 180–660 MW; each contributes 100 training and 40 validation transitions. Within each stratum, candidates are thinned to at least 60 s spacing and selected by deterministic farthest-point coverage in standardized (p_m, h_m, N_e) coordinates, without quota borrowing. The deployed dictionary is frozen at 500 points. Plant-specific training-block replay fixes $\epsilon_\kappa = 0.1$, after which the 200 time-isolated validation points are evaluated once without retuning. The public controller exports confirm the implemented value and online execution; the record-level plant replay remains a restricted enterprise asset and is not reconstructed from those exports.

Coverage monitoring combines the input z-score and 0.5th–99.5th training quantiles, the validation-set 99th percentile of posterior standard deviation, and the standardized one-step residual-rate innovation. Moderate OOD enters full-row nominal-HOCBF operation for at most ten 1 s cycles; GP operation resumes only after ten consecutive healthy cycles. A persistent coverage failure enters latched live-DCS bypass. Innovation above 5, invalid measurements or communication, non-finite/model-load faults, nominal-QP or actuator-constraint infeasibility, violation of a direct physical bound, pressure-low physical margin below 2.0 MPa, or QP/task timeout bypasses the degraded stage immediately. This nominal-HOCBF interval is a bounded transition mode without a GP coverage claim, not a certified substitute for the GP-RHOCBF mode.

4 Experimental Validation

The experiments evaluate RoCBF-SF as a real-time safety layer in an industrial DCS control loop. The validation addresses four issues: GP residual correction under model mismatch, robustness-factor calibration, sampling-budget fit, and post-retrofit execution evidence from plant historian and controller-log records.

4.1 Boiler-Turbine Benchmark and Filter Comparisons

The benchmark retains the five process states of the published boiler–turbine model and adds feedwater-flow and turbine-valve actuator states, giving the seven-state vector $x = [r_B, p_m, h_m, N_e, \tau_f, D_{fw}, u_t]^\top$. The identified feedwater and valve time constants are 15 and 6 s; the fuel-transport time constant is 30 s. The normalized command deviation is $v \in [-1,1]^3$ with physical scales $S_u = \mathrm{diag}(10,40,1)$. The enforced limits are pressure (13–24 MPa), separator enthalpy (2670–2830 kJ/kg), and generator active power around the dispatch target ($N_{\mathrm{ref}} \pm 50$ MW). All six command-level barriers have relative degree two. The HOCBF uses the sample-matched surrogate $f_s = (A_d - I)(x - x_0)/T_s$ and $g_s = B_{\mathrm{cont}} S_u$; the rollout applies its one-step forward-Euler map $x_{k+1} = x_0 + A_d(x_k - x_0) + T_s g_s v_k$. Thus, the HOCBF rows and controller-instant transition use the same drift and command matrices. Table 2 tests the $\Delta g = 0$ structural scope of Theorem 1: the rollout plant and filter share the same fixed input matrix, so mismatch enters through drift residuals. The finite rollouts do not by themselves verify the theorem's uniform residual and derivative-bound assumptions or inter-sample invariance.

Six perturbation scenarios model heat-absorption loss (S1), feedwater-induced pressure disturbance (S2), coupled state-dependent drift (S3), nonlinear fouling (S4), valve-related process drift (S5), and fuel-quality variation (S6). S5 is represented as equivalent state drift rather than as an input-gain fault, so the $\Delta g = 0$ condition is maintained. S5 and S6 also perturb the power/fuel-delay states. Scenario magnitudes are frozen before the confirmatory multi-seed sweep at the smallest screened scale that produces a nonzero 500 s unfiltered violation while retaining the equilibrium in the true HOCBF extended set.

To isolate the safety layer in the $\Delta g = 0$ validation, the fixed-proposal and HOCBF rows in Table 2 use the same seeded, bounded upstream deviation command and differ only in the downstream filter. Initial states satisfy both $h_j(x) \geq 1$ and the true-drift $\psi_{j,1}(x) \geq 0$ conditions; this oracle check is used only to define comparable evaluation starts and is unavailable to the controller. Each method-condition cell contains 50 independent 500-sample rollouts (25,000 controller samples). All GP variants use 500 scenario-specific training transitions, the common input/output vector $[p_m, h_m, N_e]^\top$, and a frozen training-input z-score transform. The QP is solved with qpax 0.1.3 in 64-bit arithmetic using row scaling, box constraints, and an acceptance requirement of solver convergence, finite primal values, and maximum normalized primal residual no greater than 10^{-6}. Rejected QP candidates revert to the live upstream proposal and are counted as fallback events. A raw-margin numerical tolerance of 10^{-2} is used only for counting binary state-limit violations; unrounded minimum margins remain in the result files.

NMPC is evaluated under the same $\Delta g = 0$ dynamics as an optimization-based reference. It solves a warm-started SLSQP problem with horizon $N = 5$, normalized deviation-input bounds $v_i \in [-1,1]$, the same six process constraints, the same sample-matched seven-state predictor, and additive disturbance correction. The output and input weights are diag($1.0,0.001,0.01$) and diag($0.01,0.01,0.01$), respectively; the disturbance-estimator gain is 0.5, and SLSQP uses maxiter=50 and ftol= 10^{-4}. NMPC uses its own receding-horizon proposal and is reported as an optimization reference rather than as a same-proposal filter ablation.

The safety tables use a sample-level violation rate at the controller instants,

$$R_{\mathrm{viol}}(\eta) = \frac{\sum_{s=1}^{N_{\mathrm{seed}}} \sum_{e=1}^{N_{\mathrm{ep}}} \sum_{t=1}^{N_{\mathrm{step}}} \mathbf{1}\left\{\min_j h_j(x_{s,e,t}) < -\eta\right\}}{N_{\mathrm{seed}} N_{\mathrm{ep}} N_{\mathrm{step}}}, \tag{13}$$

where a sample is counted once if any enforced pressure, separator-enthalpy, or power barrier is below the stated numerical tolerance η at that controller sampling instant. Table 2 uses $\eta = 10^{-2}$ in raw barrier units to avoid counting QP boundary-roundoff as physical violation; the later stress-test diagnostics use the tolerance stated in their result files or captions. This is not an episode-level probability. Counts in Fig. 2 refer only to its diagnostic rollout, whereas Table 2 aggregates controller samples across seeds and rollouts for the $\Delta g = 0$ validation.

4.2 Closed-Loop Safety and Process Response

Table 2. Primary $\Delta g = 0$ validation: sample-level constraint violation rate (%) at controller instants, computed by Eq. (13) with a raw-margin numerical tolerance of 10^{-2}. Each cell aggregates 50 independent 500-sample rollouts (25,000 controller samples). The fixed-proposal, HOCBF, and RoCBF-SF rows share the same upstream command; NMPC uses its own receding-horizon proposal. The $\epsilon_\kappa = 1$ row is an implemented endpoint test, not the recommended commissioned setting.

Method	Nom.	S1	S2	S3	S4	S5	S6
Unfiltered fixed upstream proposal	0.00	64.01	25.75	39.19	61.88	11.65	58.06
HOCBF, no GP (all $m = 2$)	0.00	64.01	25.75	39.19	61.88	11.65	58.06
NMPC ($N = 5$)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
RoCBF-SF, tune-selected ($\epsilon_\kappa = 0$)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Full implemented-margin endpoint ($\epsilon_\kappa = 1.0$)	0.00	64.01	25.75	39.19	61.88	11.65	58.06

Table 2 reports the drift-only $\Delta g = 0$ validation. Across the six mismatch conditions, the fixed upstream proposal and HOCBF without GP correction each record 65,134 violating samples among 150,000 controller samples. The identical counts show that structurally correct command-level HOCBF rows alone do not compensate the imposed drift residuals in these rollouts. The S3 tune/test procedure selects the smallest passing value, $\epsilon_\kappa = 0$, and the resulting GP-mean RoCBF-SF row records 0/175,000 observed violations across the nominal case and all six mismatch conditions. Each condition uses its own frozen scenario-specific GP; the controller does not identify the true scenario online. NMPC also records 0/175,000 violations and no solver failure. The comparison therefore does not establish universal superiority over MPC; RoCBF-SF's deployment distinction is its one-step downstream projection of an existing coordinated-controller command.

The full implemented-margin row shows why ϵ_κ must be commissioned rather than set mechanically to its largest value. It remains feasible in the nominal condition, but rejects 149,800/150,000 QPs across the six mismatch conditions and consequently reproduces the upstream proposal's 65,134 violations. This outcome does not contradict Theorem 1: the numerical endpoint does not satisfy the theorem's full-row robust-QP-feasibility condition, and the finite benchmark does not independently establish its uniform residual event or derivative bounds.

Table 3 makes that feasibility mechanism explicit for S3. The experiment records every QP attempt, rejected candidate, fallback, intervention, and normalized residual. The tune-selected GP-mean row accepts every QP and records no state-limit violation; the full implemented-margin endpoint rejects all 25,000 QPs and reverts to the upstream proposal throughout.

Table 3. ***S3 solver and intervention diagnostics for the primary $\Delta g = 0$ validation. Each row contains 25,000 controller samples. QP rejection requires non-convergence, a non-finite candidate, or normalized primal residual above 10^{-6}; the live upstream proposal is used on a rejected cycle. NMPC uses its own horizon optimization and is therefore omitted from the projection-intervention comparison.***

S3 method	Violation	QP rejected	Fallback	QP intervention
HOCBF, no GP	9798 of 25,000 (39.19%)	0 of 25,000	0 of 25,000	0.00%
RoCBF-SF, $\epsilon_\kappa = 0$	0 of 25,000	0 of 25,000	0 of 25,000	41.32%
Full implemented-margin endpoint	9798 of 25,000 (39.19%)	25,000 of 25,000	25,000 of 25,000	0.00%

The results identify a mechanism hierarchy within the tested benchmark: the actuator-augmented HOCBF supplies the required relative-degree-two command constraints, GP residual correction produces the observed mismatch compensation, and finite-sample calibration prevents unsupported uncertainty tightening from destroying QP feasibility. The following diagnostics separate the time-domain mechanism, residual-model evidence, data-quality gate, and tune/test calibration result.

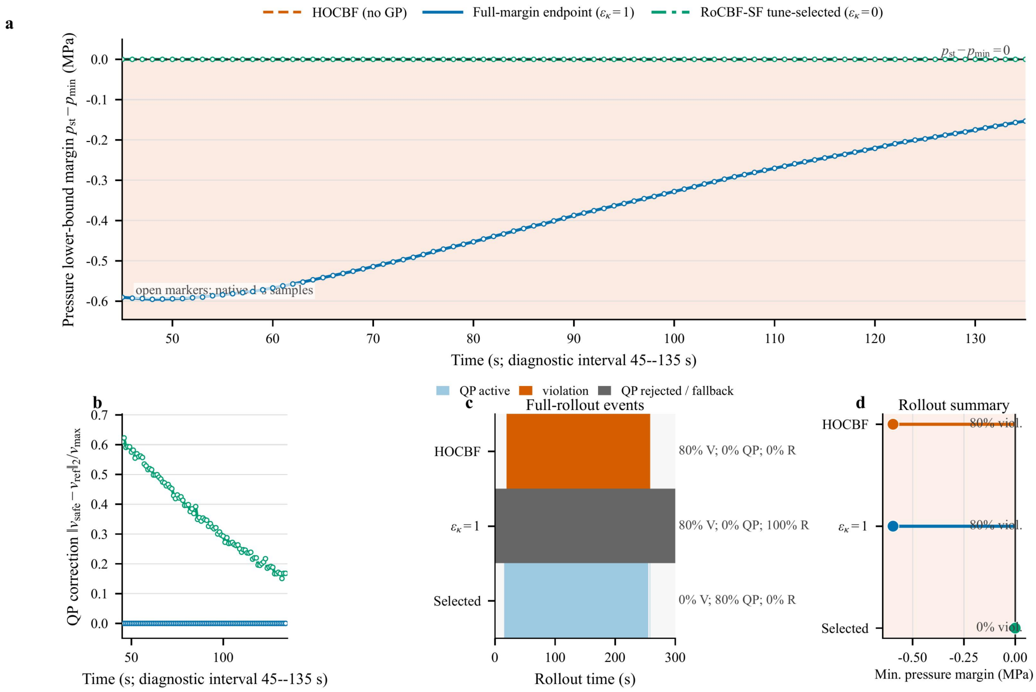

Figure 2. Closed-loop pressure-bottleneck mechanism for representative seed 3 under the drift-only S3 perturbation. **(a,b)** Separator-pressure lower-bound margin and normalized QP correction over the 45–135 s diagnostic interval. The shaded region marks violation of the lower pressure constraint. **(c)** Full 300 s event record, distinguishing accepted QP intervention, state-limit violation, and rejected-QP fallback. **(d)** Full-rollout minimum pressure margin and sample-level violation rate; the symmetric-log horizontal scale resolves the tune-selected near-boundary trajectory while retaining the no-GP excursion. Lines in (a,b) are interpolated on a 0.1 s display grid only for visual continuity; open markers are the native 1 s controller samples, and all statistics use native samples and the 10^{-2} counting tolerance defined in Eq. (13).

Figure 2 shows the pressure bottleneck directly in a representative S3 rollout. HOCBF without GP correction produces 239/300 violating samples and a minimum lower-pressure margin of -0.59553 MPa. The full implemented-margin endpoint rejects every QP candidate in this rollout and therefore follows the fallback proposal, producing essentially the same pressure excursion. The tune-selected mean-only setting ($\epsilon_\kappa = 0$) accepts all 300 QPs, intervenes on 241 samples, and records 0/300 violations under the stated counting tolerance; its unrounded minimum pressure margin is -4.94×10^{-5} MPa. The correction and event panels therefore attribute the change to accepted GP-mean-corrected QP intervention rather than to a different upstream proposal. This single-trajectory diagnostic illustrates the mechanism and is not used in place of the five-seed aggregate in Tables 2 and 3.

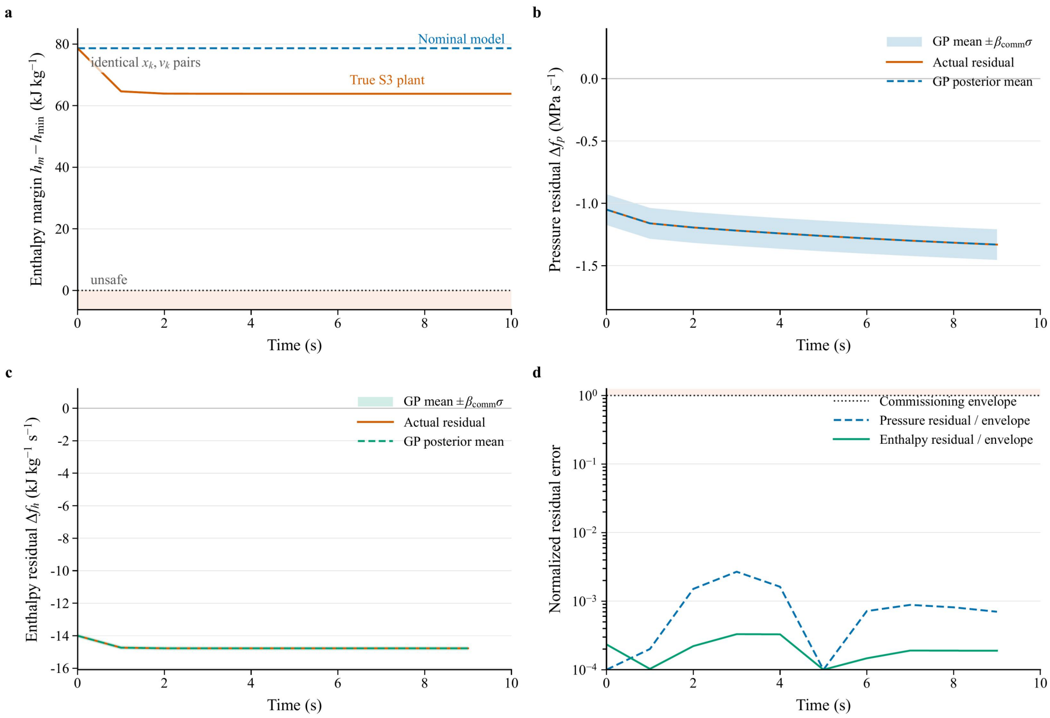

Figure 3. Direct model-mismatch diagnosis under the drift-only S3 perturbation. **(a)** Nominal and true one-step enthalpy margins from identical state–input pairs. **(b,c)** Pressure and enthalpy residual rates compared with the scenario-specific GP posterior mean and commissioning envelope ($\mu_{GP} \pm \beta_{comm}\sigma_{GP}$, $\beta_{comm} = 12.262$, $N = 500$). **(d)** Posterior residual error normalized by that implemented envelope; values below one lie inside the plotted diagnostic envelope, not a separately established uniform GP confidence set.

Figure 3 makes the model mismatch explicit. At the first controller instant, the same state and zero-deviation input give nominal and true next-step separator pressures of 16.546 and 15.582 MPa, respectively; the corresponding nominal and true lower-enthalpy margins are 78.628 and 64.628 kJ/kg. Along the diagnostic trajectory, the realized residual rates range from -1.497 to -1.050 MPa/s for pressure and from -14.777 to -14.000 kJ kg^{-1} s^{-1} for enthalpy. The GP posterior mean tracks this structured discrepancy, and the normalized posterior error remains below 0.003 inside the implemented commissioning envelope. The filter therefore corrects the learned drift residual before applying any nonzero robustness multiplier.

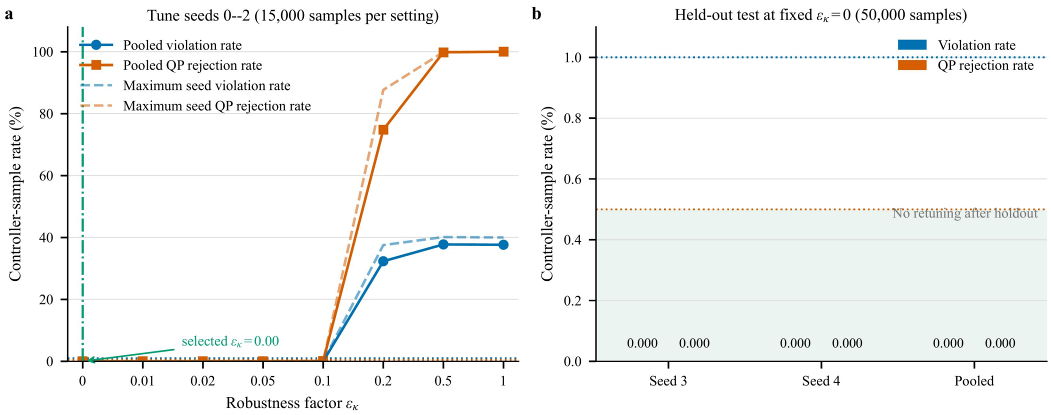

Figure 4. Tune/test calibration of ϵ_κ under S3. **(a)** Pooled and maximum-per-seed violation and QP-rejection rates for tune seeds 0–2 (15,000 native controller samples per setting). The predeclared rule selects the smallest value with pooled and per-seed violation below 1% and QP rejection below 0.5%. **(b)** Results for held-out seeds 3–4 after fixing the selected $\epsilon_\kappa = 0$ setting; no retuning used the 50,000 holdout samples.

Figure 4 separates calibration from confirmation. On tune seeds 0–2, $\epsilon_\kappa = 0$, 0.01, 0.02, and 0.05 each give 0/15,000 observed violations and no rejected QP; the predeclared smallest-passing rule therefore selects the mean-only endpoint $\epsilon_\kappa = 0$. At $\epsilon_\kappa = 0.1$, no violation is observed but 13/15,000 QPs are rejected. Values of 0.2 and above fail the feasibility gate, and the full endpoint rejects all 15,000 tune-set QPs. After fixing $\epsilon_\kappa = 0$, held-out seeds 3–4 give 0/50,000 observed violations and 0/50,000 rejected QPs without retuning. The zero-count Wilson upper value reported in the Supplemental Material is only a descriptive Bernoulli reference because adjacent controller samples are serially correlated. The result is a finite-sample selection outcome, not proof that the violation probability is zero.

4.3 Robustness-Factor Commissioning

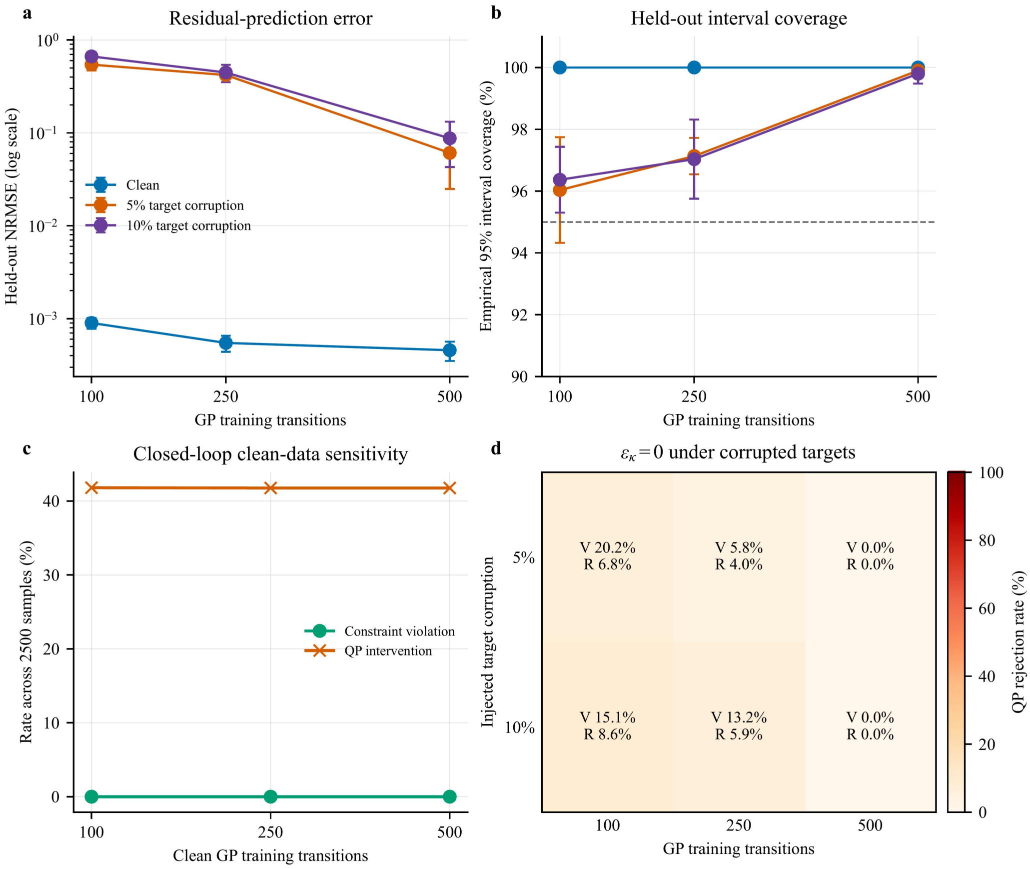

Figure 5. Controlled GP training-data sensitivity under S3. **(a,b)** Held-out residual NRMSE and empirical 95% interval coverage for 100, 250, and 500 selected transitions. Target corruption replaces a fixed training fraction with signed three-standard-deviation offsets. **(c,d)** Closed-loop violation and QP-rejection diagnostics under clean and corrupted training data. Each point aggregates five 500-sample rollouts. The 5% and 10% levels are synthetic fault-injection tests, not estimates of plant-data quality.

Figure 5 tests the GP commissioning gate rather than treating every fitted model as deployable. Across 100, 250, and 500 clean training transitions, held-out NRMSE decreases from 0.00090 to 0.00046, empirical interval coverage is 100%, and the tune-selected $\epsilon_\kappa = 0$ setting gives 0/2500 violations and no QP rejection at each size. Synthetic target corruption exposes a dictionary-size dependence: the 100- and 250-point fits produce 5.76–20.24% violation and 4.04–8.60% QP rejection, whereas the 500-point fits retain 0/2500 observed violations and no QP rejection at both tested corruption levels, with 99.8–99.9% coverage. The result supports quantity, coverage, and closed-loop feasibility as joint pre-enable gates; it does not estimate plant-data corruption or establish universal robustness to bad labels. The complete aggregation is reported in the Supplemental Material.

4.4 Real-Time Computation

The deployment configuration specifies an AMD EPYC Embedded 4565P processor, 64 GB (2×32 GB) DDR5-5600 ECC memory, openEuler 24.03 LTS, Python 3.11, qpax 0.1.3 for the online QP, SciPy 1.17.1, Kubernetes workload management, and Modbus TCP data exchange with the DCS. The online controller executes every 1 s. The production-validation files are controller exports at 5 s spacing; therefore, counts reported from those files refer to exported records rather than every internal controller cycle.

The implemented benchmark filter and NMPC reference both fit within the 1 s budget; scenario-level timing and GP storage/training complexity are reported in the Supplemental Material. Field execution time is the deployment-relevant measurement and is reported with the confirmed controller exports in Section 4.5.

4.5 Measured Plant Evidence: Envelope, Matched Windows, and Controller Logs

The simulation benchmark establishes the closed-loop mechanism: GP residual correction closes most of the model mismatch, ϵ_κ must be calibrated by uncertainty structure, and the compiled filter satisfies the 1 s sampling budget. Plant data then connect the same design to measured operation. Historian records define the operating envelope, load-matched windows compare pre- and post-retrofit response, and confirmed original plant-controller exports document both routine full-QP execution and a bounded reduced-QP recovery mode under measured operating conditions.

Figure 6 uses a bounded historian snapshot from the 660 MW unit as an operating-envelope check. The grey envelope uses an approximately 24 h loaded operating window from 1 July 2026 02:33 to 2 July 2026 02:28 local time ($n = 288$ at 300 s spacing). The colored overlays place three post-retrofit low-to-mid-load historian windows within that measured load–pressure range. This record documents measurement availability and operating variability; the original 5 s historian records in Fig. 7 provide the dynamic resolution for retrofit comparison.

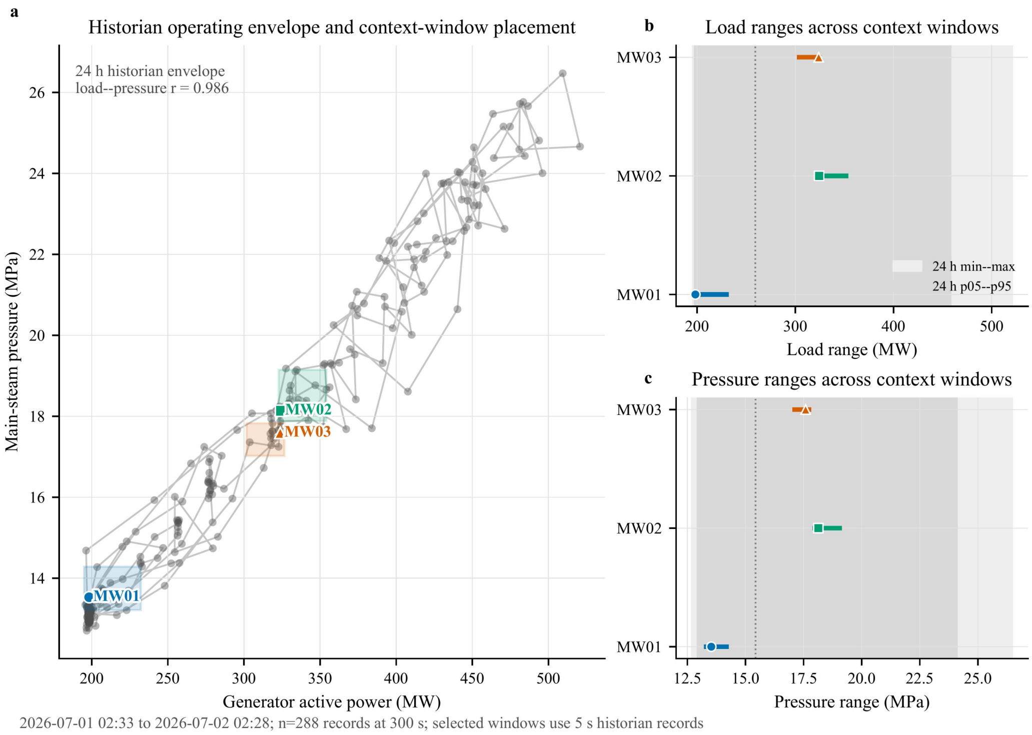

Figure 6. Plant-historian operating-envelope check from the 660 MW ultra-supercritical unit. **(a)** Five-minute historian samples define the measured load–pressure envelope over an approximately 24 h loaded interval; colored boxes and markers locate three two-hour post-retrofit low-to-mid-load historian windows. **(b,c)** The same windows are shown as load and pressure ranges against the 24 h min–max and p05–p95 envelopes. The figure places these historian windows within measured plant operation.

During this interval, generator active power varied from 195.7 to 520.8 MW and main-steam pressure ranged from 12.70 to 26.47 MPa. This 24 h range is a historian operating-envelope measurement, not the safety-filter constraint envelope. The three colored overlays in Fig. 6 are low-to-mid-load historian context windows (MW01–MW03), whose main-steam-pressure ranges are 13.21–19.15 MPa. They do not supply controller-internal QP fields. Load was strongly correlated with pressure ($r = 0.986$) and fuel flow ($r = 0.980$), confirming pressure–combustion coupling during load following. The MW04–MW06 controller exports analyzed below provide the separate execution evidence at 480.7–629.7 MW (72.8–95.4% of nameplate power). Taken together, the two record types span measured operation from 195.7 to 629.7 MW (29.7–95.4% of nameplate), but they do not constitute continuous controller-internal performance validation at every load through 660 MW.

The unit chronology permits a load-matched commissioning comparison. The 660 MW unit used traditional PID before 21 May 2026, was shut down from 21 May to 24 June 2026, and restarted with the proposed RoCBF-SF-based strategy after 24 June 2026. The outage included extensive routine plant maintenance in addition to integration of the proposed algorithm. We screened 300 s historian records from 2024–2025 and 25 June–5 July 2026. Candidate two-hour windows contain 24 consecutive samples, advance every 30 min, and retain only load in 195–700 MW with finite pressure, setpoint, and fuel-flow values. A global linear-assignment step forms one-to-one pairs without replacement using mean-load difference, 24-point load-profile RMSE, load-range difference, and fuel/load difference. Calipers are 10, 30, and 40 MW and 0.08, respectively. This protocol yields 512 pairs from 14,675 pre-retrofit and 517 post-retrofit candidates. Median mean-load difference is 1.32 MW and median profile RMSE is 15.37 MW. Cohort median pressure-residual standard deviation decreases from 0.523 to 0.502 MPa (3.93%), whereas median absolute-error p95 is essentially unchanged (1.082 versus 1.085 MPa). Because the windows overlap, these cohort quantities and the post-day cluster bootstrap reported in the Supplemental Material are descriptive rather than independent-sample inference. Figure 7 is a separate native-5 s load-matched diagnostic, not a selected member of the 300 s cohort; its source timestamps and metric JSON are preserved independently in the evidence package. Concurrent routine maintenance also means that the response comparison is observational evidence associated with the documented retrofit, not a standalone causal estimate. Confirmed original plant-controller exports provide the separate evidence that RoCBF-SF executed after restart.

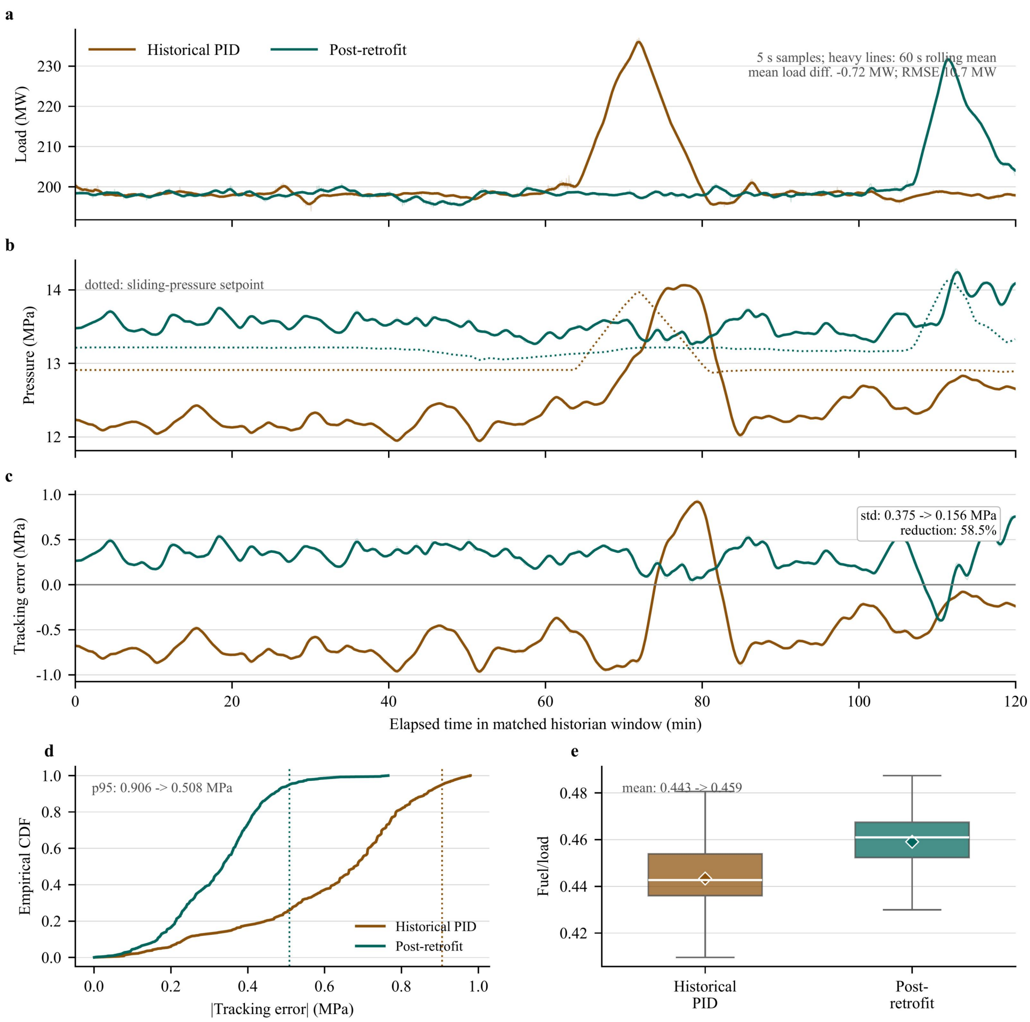

Figure 7. Matched plant-historian retrofit-window diagnostic from the 660 MW ultra-supercritical unit. The historical PID window is 5 November 2025 11:00–12:59:55 local time; the post-retrofit window is 25 June 2026 12:30–14:29:55 local time. Both windows contain 1440 original 5 s historian samples. **(a)** Generator active-power trajectories used for matching; the mean-load difference is -0.72 MW and the load-profile RMSE is 10.7 MW. **(b)** Main-steam pressure and the sliding-pressure setpoint. **(c)** Pressure tracking error, $p_{\text{main}} - p_{\text{set}}$; the sample-level standard deviation decreases from 0.375 to 0.156 MPa. **(d)** Empirical CDF of absolute pressure tracking error; dotted lines mark the 95th percentile, which decreases from 0.906 to 0.508 MPa. **(e)** Fuel/load distribution for the combustion-side operating context; boxes show median, IQR, and 1.5×IQR whiskers, and diamonds show means. Light traces are original samples and heavy traces are 60 s rolling means for readability; all reported metrics use the original 5 s samples.

Figure 7 reports a high-resolution matched pair. The two-hour windows have mean load within 1 MW and load-profile RMSE of 10.7 MW. Pressure tracking-error standard deviation decreases by 58.5% after retrofit, from 0.375 to 0.156 MPa, and the 95th-percentile absolute error decreases from 0.906 to 0.508 MPa. The fuel/load proxy is higher after retrofit (0.443 to 0.459), so the comparison was not obtained under lower combustion-side demand. The load-matched result documents improved pressure response after the retrofit under comparable load movement. Because routine outage maintenance occurred concurrently, this comparison does not by itself attribute the entire improvement to RoCBF-SF; the controller exports independently verify online execution of the proposed filter. The plant evidence complements the controlled model-mismatch tests in Figs. 2–3.

The same 5 s historian query includes main/reheat temperature, air–coal ratio channels, mode/fallback indicators, and turbine-valve demand. Main-steam temperature variability decreases from 2.245 to 0.915C, while reheat-temperature variability and air–coal ratio dispersion increase, showing that the post-retrofit interval is not uniformly easier. These historian diagnostics are then paired with the post-retrofit controller-log export in Fig. 8; the historical PID window is not assigned QP/CBF fields because no online safety filter was running before retrofit.

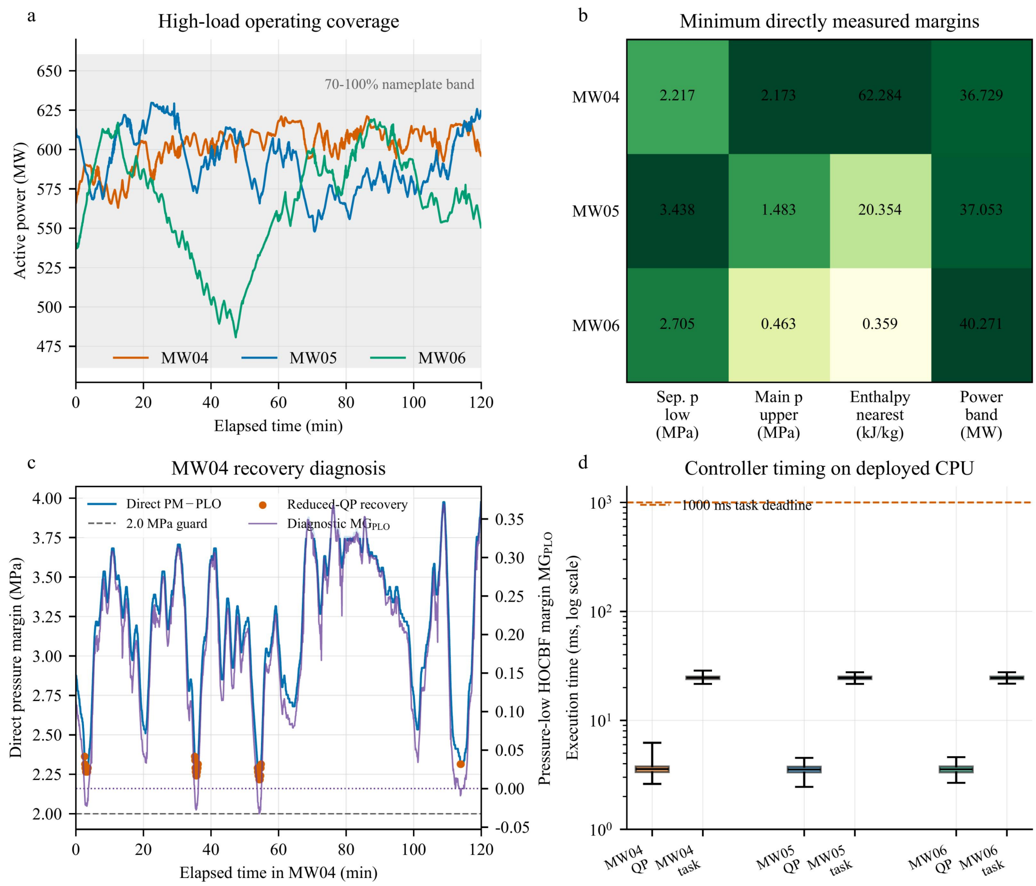

Figure 8. High-load plant-controller evidence from three post-retrofit two-hour windows, each containing 1440 records exported at 5 s spacing from a controller executing at 1 s. **(a)** Generator active power; the shaded band denotes 70–100% of the 660 MW nameplate rating. **(b)** Minimum directly measured margins to the separator-pressure lower bound, main-steam-pressure upper replay bound, nearest separator-enthalpy bound, and ±50 MW power band. Cell shading is normalized within each column; annotations retain physical units. **(c)** MW04 recovery diagnosis: the directly measured separator-pressure margin remains above the 2.0 MPa guard while the pressure-low HOCBF diagnostic row becomes negative; orange markers denote the 43 one-retry reduced-QP records. **(d)** QP and total task execution times on the deployed CPU; whiskers span the observed ranges and the dashed line marks the 1000 ms task deadline.

Figure 8 supplies the high-load execution evidence requested for the revision. MW05 and MW06 contain 2880 records with online automatic operation, full-QP status optimal, no fallback or saturation flag, positive logged HOCBF margins, and no direct pressure, enthalpy, or power-bound exceedance.

MW04 contains 1397 optimal and 43 optimal_recovered records. In those 43 records, the original pressure-low HOCBF row has $MG_{PLO} \in [-0.03266, -0.00561]$, while the directly measured margin remains $PM - PLO \in [2.21654, 2.36306]$ MPa and the other five exported HOCBF margins remain positive. The controller therefore removes only the whitelisted pressure-low row and solves one reduced-QP retry. These cycles establish feasibility of the reduced problem, not of the original six-row QP, and the certificate in Theorem 1 does not apply to them. One fuel-saturation flag occurs in MW04; no direct physical bound is exceeded.

Across MW04–MW06, all 4320 records satisfy the logged 1000 ms task deadline. QP time has median/p95/maximum values of 3.557/4.095/6.234 ms, and total task time has corresponding values of 24.564/26.183/28.665 ms. The field robustness factor is fixed at $\epsilon_\kappa = 0.1$ after plant-specific training-block replay and one-pass time-isolated validation; it is not justified by numerical equality with the benchmark S3 calibration result. Record-level plant replay remains a restricted enterprise asset, whereas the public controller exports independently confirm the implemented value and online execution. The evidence index keeps these three controller exports separate from the MW01–MW03 low-to-mid-load historian context used in Fig. 6.

Table 4. High-load post-retrofit controller-log summary from the 660 MW ultra-supercritical unit. Each row contains 1440 records at 5 s export spacing from a controller executing at 1 s. Direct margins are minimum values over the complete window. $p_{\mathrm{sep,lo}}$ is $PM - PLO$; $p_{\mathrm{main,hi}}$ is $PST\_HI - p_{\mathrm{main}}$, where PST_HI is a main-steam-pressure replay bound rather than a separator-pressure limit. “None” in the status column denotes no reduced-QP recovery and no saturation flag.

Window	Local time	Load (MW)	Min. $p_{\mathrm{sep,lo}}$ (MPa)	Min. $p_{\mathrm{main,hi}}$ (MPa)	Min. h margin (kJ/kg)	QP/recovery status	QP/task p95 (ms)
MW04	29 Jun 2026 19:00–20:59:55	563.0–621.2	2.217	2.173	62.284	1397 optimal; 43 recovered; 1 fuel-sat.	4.195 / 26.238
MW05	30 Jun 2026 14:00–15:59:55	547.9–629.7	3.438	1.483	20.354	1440 optimal; none	4.074 / 26.139
MW06	26 Jun 2026 23:00–00:59:55	480.7–619.3	2.705	0.463	0.359	1440 optimal; none	4.062 / 26.159

Table 4 separates the two high-load execution roles rather than pooling them into a uniformly clean claim. MW05–MW06 show routine full-QP operation; MW04 exposes the recovery mechanism and its certificate boundary. Together, Figs. 6–8 separate three evidence types: controlled simulation and GP residual diagnostics establish the model-mismatch mechanism, matched historian windows compare plant response under comparable load movement, and plant-controller exports verify the deployed execution and recovery paths.

4.6 Stage-Gated Commissioning Protocol

RoCBF-SF is intended as a **commissioning-calibrated safety filter**. The result sequence implies a staged deployment logic: simulation and replay establish the residual model and margin envelope, supervised tests verify QP/fallback behavior, and production logs verify execution after enablement.

Table 5 gives a compact main-text version of this protocol; the supplemental material provides the detailed evidence inventory and window-level status.

Table 5. Compact stage-gated commissioning protocol for deploying RoCBF-SF as a safety-filter overlay.

Stage	Gate	Minimum evidence	Decision rule
Offline	Residual and margin calibration	Held-out GP coverage, representative historian/replay samples, and local ϵ_{κ} sweep with violation, intervention, and infeasibility metrics	Use a field-calibrated frozen GP whose training domain covers the commissioned operating envelope; select the smallest ϵ_{κ} that meets the local safety gate without QP infeasibility.
Supervised	Pre-enable validation	Time-isolated replay, protection-limit review, alarm/fallback tests, and coverage across the five fixed load strata	Enable online authority only after GP, QP, fallback, and alarm logic pass supervised tests.
Online	Execution logging	Aligned records for v_{prop}, v^*, reconstructed actuator command u_k, QP status, active CBF rows, $\sigma_{\text{total}}(x)$, fallback, saturation, and solve time	Treat historian traces as plant-response diagnostics; use controller logs such as Fig. 8 to support safety-filter execution.
Online	Drift and boundary monitoring	Drift in residual coverage, intervention rate, active enthalpy margin, fallback frequency, or repeated QP relaxation	Recalibrate the GP and repeat the local sweep; if the envelope boundary is reached, fall back to conservative operation instead of increasing ϵ_{κ}.

Margin candidates are evaluated only in offline replay or supervised commissioning windows. The benchmark uses the smallest tune-seed value that passes the predeclared violation and QP-rejection gates and then tests that value without retuning. The plant value $\epsilon_{\kappa} = 0.1$ is a separate operating point obtained from the time-separated plant replay; it is not transferred numerically from S3. During operation, moderate coverage loss enters full-row nominal-HOCBF mode for at most ten cycles, whereas persistent coverage loss or a hard fault returns authority to the live DCS command and latches bypass. The $\Delta g = 0$ assumption limits the formal certificate to drift-dominated uncertainty; actuator-gain uncertainty requires additional modeling.

The Supplemental Material provides the full tune/test table, GP data-quality diagnostics, the sampled-data claim boundary, field commissioning and fallback details, production-evidence traceability, computation-time data, and the proof of Theorem 1.

5 Conclusion

This paper developed RoCBF-SF as a commissioning-calibrated GP-HOCBF safety-filter overlay for the boiler-turbine coordinated control loop of an ultra-supercritical power unit. The filter leaves the upstream controller in place and projects its proposed command through pressure, separator-enthalpy, and power constraints using GP drift-residual correction, compositional HOCBF uncertainty propagation, and a real-time QP.

The seven-state, drift-only $\Delta g = 0$ benchmark separates barrier structure, residual learning, and margin selection. Across six mismatch conditions, HOCBF without GP correction records 65,134/150,000 violating samples. The predeclared calibration rule selects the mean-only endpoint $\epsilon_\kappa = 0$; this RoCBF-SF setting and the NMPC reference each record 0/175,000 observed violations in the primary sweep, and the fixed filter setting records 0/50,000 violations and no QP rejection on held-out seeds. The equality with NMPC precludes a claim of universal safety superiority; the practical distinction is the filter's compatibility with an existing coordinated controller. Conversely, the implemented $\epsilon_\kappa = 1$ endpoint loses feasibility in nearly all mismatch-scenario QPs under the tested actuator limits. The calibrated settings remain finite-sample operating points. The formal forward-invariance result requires $\epsilon_\kappa = 1$, an independently valid simultaneous residual event and compositional derivative bound, $\Delta g = 0$, and full-row robust-QP feasibility; the implemented multiplier does not establish those conditions by itself.

Plant historian and controller records provide complementary measured-operation evidence. The 512-pair historian cohort shows only a small median pressure-dispersion change with a broad post-day cluster-bootstrap interval, whereas the separate native-5 s pair shows a larger response change under comparable load movement. Both are observational because routine maintenance accompanied the retrofit. Three confirmed high-load plant-controller exports verify the proposed execution path over 480.7–629.7 MW. Direct pressure, enthalpy, and power margins remain positive over all 4320 exported records; MW05–MW06 use the full QP throughout, and MW04 records 43 guarded reduced-QP recoveries. The observed total task time remains below 28.665 ms against the 1000 ms deadline. These recovered cycles are excluded from the full-row certificate claim.

Future work should extend time-separated commissioning coverage, evaluate actuator-gain uncertainty beyond the current $\Delta g = 0$ formulation, and develop a sampled-data certificate for the 1 s zero-order-hold implementation. Further field validation should also retain the current separation between historian response evidence, controller-execution evidence, and full-row certificate conditions.

Statements and Declarations

Ethical considerations

Not applicable. This study did not involve human participants, human data, human tissue, or animal experiments.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data availability

The source code, simulation scripts, benchmark results, plotting scripts, derived plant-historian metrics, and bounded controller-export excerpts are available in the GitHub repository. Complete raw plant records are proprietary enterprise assets. Qualified researchers may request restricted access from the corresponding author, subject to data-owner approval and an executed data-use agreement.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used Grammarly for English grammar checking. The authors reviewed and edited the manuscript and take full responsibility for its content.

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